Crimson Stambaugh

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Education

University of Washington Seattle, WA
M.S. Applied and Computational Mathematics September 2024 – June 2025
GPA: 3.75

Southern New Hampshire University Manchester, NH
B.A. Mathematics May 2020 – May 2024
GPA: 3.992, *Outstanding Student Award for Mathematics*

Publications

Mixed Density Diffuser: Efficient Planning with Non-Uniform Temporal Resolution
Crimson Stambaugh, Rajesh Rao
ESSAN, 2026

Multi-modal Natural Intelligence through Active Predictive Coding
Jeffrey Duan, Vishwas Sathish, *Crimson Stambaugh* Rajesh Rao
AAAI "Towards Multi-Modal Natural Intelligence" Workshop, 2026

Professional Research Experience

University of Washington Seattle, WA
Research Scientist, Supervisors: Rajesh Rao, Jaemin Cho 06/2025 – present

⋆ **Video Action Models**
- Optimizing few-shot generalization in video-action models for real world application
- Training latent video diffusion models for world modeling

⋆ **Diffusion-Based Offline Reinforcement Learning**
- Developed novel non-uniform temporal resolution diffusion planner
- Achieved state-of-the-art offline reinforcement learning performance on D4RL benchmark

⋆ **Multi-modal Active Predictive Coding**
- Applied novel sensory-motor neuroscience theory for meta reinforcement learning
- Built novel meta-learning algorithms to explore sensory and motor activity interactions in the neocortex.

Projects

University of Washington Seattle, WA

⋆ **VLA Benchmark**
- Used ManiSkill framework to build Vision-Language-Action skill generalization benchmark
- Worked in a four-person team to develop evaluations and establish baseline performance with the RT-1 Robotics Transformer

- Supervised by Jiafei Duan

Southern New Hampshire University Manchester, NH

⋆ **GANs for Image-to-Image Translation**

- Constructed, trained, applied, and tested conditional GANs (cGANs) and cycle-consistent adversarial nets (CycleGANs) in image-to-image translation applications using TensorFlow
- Analyzed relevant academic literature on the invention and evolution of GANs, cGANs, and CycleGANs
- Supervised by Dr. Adam Gilbert

⋆ **Unmanned Aerial Vehicle (UAV) Wide Area Search Algorithm**

- Collaborated with a BAE research scientist to develop a wide area search algorithm to decompose a search area into tiles & construct an efficient UAV route plan
- Presented results to BAE researchers, SNHU faculty, and fellow students
- Supervised by Dr. Christine Caples

MIT Beaver Works Summer Institute Kwajalein, Marshall Islands

⋆ **miniRACECAR Project/ Serious Game Development**

- Programmed a car to follow and prioritize colored lines, and park in front of a cone using computer vision principles
- 1st place in the team final competition of 2020
- Individually earned the Top MIT Beaver Works Coder Award
- Supervised by Dr. Sarah Willis

Teaching Experience

Southern New Hampshire University Manchester, NH

Learning Fellow/Class Embedded Tutor November 2022 – April 2024

- Explained calculus, physics, statistics, and linear algebra concepts clearly and concisely during drop-in tutoring sessions and exam reviews.
- Analyzed pedagogical experiment results and discussed course plans in a weekly huddle with a three-person Learning Fellows team.
- Received Learning Fellow of the Month award.

MIT Beaver Works Summer Institute Kwajalein, Marshall Islands

Teaching Assistant July 2022 – August 2022

- Part of a four-person team, teaching Python and computer vision concepts to 18 high-school students.
- Overcame language and cultural barriers to instruct 12 Marshallese students, for whom English was a second language.

Relevant Coursework

Graduate: Deep Learning, Optimization, Advanced Random Processes, Scientific Computing, Probability, Chaos and Dynamical Systems
Undergraduate: Statistical Learning, Regression Analysis, Numerical analysis, Real Analysis, Neuroscience, Artificial Intelligence

Skills

Programming: Python, R, MATLAB, LaTeX
ML Tools: PyTorch, scikit-learn, NumPy, pandas